Mathematical examination of the holographic-brain paper

Independent reconstruction, counterexamples, and unresolved derivations

Technical examination of Nishiyama–Tanaka–Tuszynski–Tsenkova (2024)

5 November 2026

Paper: *Holographic Brain Theory: Super-Radiance, Memory Capacity and Control Theory*, *Int. J. Mol. Sci.* **25** (2024), 2399; DOI 10.3390/ijms25042399.
Source equation numbers (1)–(11), (A1)–(A91) retained in annotations.
Protocol: SOURCE equation → direct calculation → condition or counterexample → residual.

0. The declared mathematical system

Source: §1; Appendix A

$$c = \hbar = 1, \quad \eta_{\mu\nu} = \mathrm{diag}(1,-1,-1,-1), \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad D_i = \partial_i - \mathrm{i}eA_i.$$

$$\boxed{\mathcal{L}_A = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + \phi^*\left(\mathrm{i}\partial_t + eA^0 + \frac{D_iD_i}{2m} - U(x)\right)\phi.} \tag{A1}$$

$\phi \in \mathbb{C}$ nonrelativistic charged scalar; A_μ electromagnetic potential; e, m, U model inputs.

$$\underbrace{\text{complex field } \phi}_{\text{chosen}} \longleftrightarrow \underbrace{\text{water conformational states}}_{\text{not derived}}, \qquad \underbrace{A_\mu}_{\text{chosen}} \longleftrightarrow \underbrace{\text{microtubule emission}}_{\text{not measured}}.$$

Object of examination: equations and consequences, not the biological names.

1. Maxwell constraints: do the charges fit the gauge?

Source: A2–A7; Appendix A, pp.13–14

$$\mathbf{E} = -\partial_t\mathbf{A} - \nabla A^0, \quad \mathbf{B} = \nabla \times \mathbf{A},$$

$$\nabla \cdot \mathbf{E} = \rho, \qquad \nabla \cdot \mathbf{B} = 0,$$
$$\nabla \times \mathbf{E} = -\partial_t\mathbf{B}, \qquad \nabla \times \mathbf{B} - \partial_t\mathbf{E} = \mathbf{J}.$$

Here electromagnetic unit factors are absorbed into e, J, ρ in the paper's natural-unit convention. The independent $\nu = 0$ Maxwell equation is Gauss's constraint.

$$\underbrace{A^0 = 0, \quad \nabla \cdot \mathbf{A} = 0}_{\text{both imposed after (A1)}} \Longrightarrow \nabla \cdot \mathbf{E} = -\partial_t \nabla \cdot \mathbf{A} - \nabla^2 A^0 = 0.$$

$$\underbrace{\partial^\nu F_{\nu 0} = J_0}_{\text{paper (A7)}}, \qquad \underbrace{J_0 = -e|\phi|^2}_{\text{paper (A6)}} \Longrightarrow \boxed{0 = -e|\phi|^2.}$$

$$\int |\phi|^2 \mathrm{d}^3x = N > 0, \quad e \neq 0 \quad \Longrightarrow \quad \boxed{\text{contradiction.}} \text{ [FAIL AS WRITTEN]}$$

Independently from Maxwell: the paper's (A3)–(A7) yields exactly the same constraint. No external law is required.

Possible repair, not performed by paper:

$$\nabla \cdot \mathbf{A} = 0, \quad \nabla^2 A^0 = -\rho \quad (A^0 \text{ not fixed to } 0).$$

Alternatively, specify and bound a transverse-field approximation. If an effective neutralizing background $\rho_b(\mathbf{x})$ is proposed, then $\rho + \rho_b = 0$ under the exact combined gauge: a fixed ρ_b forces static ρ at each point. A time-dependent background requires its own continuity and dynamics.

Owed: A consistent Gauss constraint, source charge/current, and a controlled bound on any dropped longitudinal field.

2. A 1D source paired with a 2D field

Source: A8–A13; pp.14–15

$$\phi(t, \mathbf{x}) = \sum_n b_n(t)\varphi_n(\mathbf{x}), \qquad \left[-\frac{\nabla^2}{2m} + U + \frac{e^2\mathbf{A}^2}{2m}\right]\varphi_n = E_n\varphi_n. \tag{A8–A9}$$

$$\int |\phi|^2 \mathrm{d}^3x = \sum_n |b_n|^2 = N \quad \text{if } \int \varphi_n^* \varphi_m \mathrm{d}^3x = \delta_{nm}.$$

Orthonormality is necessary if this identity must hold for every amplitude vector b. If $\mathbf{A} = \mathbf{A}(t, \mathbf{x})$, the displayed eigenoperator in (A9) depends on t. A fixed $\varphi_n(\mathbf{x})$ then requires an approximation; an instantaneous basis $\varphi_n(t, \mathbf{x})$ instead produces additional $\langle \varphi_m | \partial_t \varphi_n \rangle$ terms. [NOT SHOWN]

$$|\partial_3 \varphi_n| \gg |\partial_1 \varphi_n|, |\partial_2 \varphi_n|, \quad A_1 = A_2 = 0, \quad \nabla \cdot \mathbf{A} = 0 \Longrightarrow \partial_3 A_3 = 0.$$

$$\boxed{\begin{aligned}\left(\partial_t^2 - \partial_1^2 - \partial_2^2\right) A_3 + \frac{e^2|\phi|^2}{m} A_3 \\ = -\frac{\mathrm{i}e}{2m}(\phi^* \partial_3 \phi - \phi \partial_3 \phi^*).\end{aligned}} \tag{A11}$$

A_3 has no x^3 dependence, yet ϕ and the right-hand side may have x^3 dependence. Equality pointwise requires compatible x^3 dependence or an explicitly defined longitudinal averaging/projector. The paper subsequently uses spatial averages, but the pointwise transition is not established. [CONDITIONAL]

Owed: Longitudinal boundary conditions and an explicit projection/averaging operation if this is an averaged equation.

3. Two-level reduction and the resonant field

Source: A12–A35; pp.14–17

$$b_n(t) = e^{-\mathrm{i}E_n t}\chi_n(t), \quad n \in \{0, 1\}, \quad \Omega = E_1 - E_0, \quad |\chi_0|^2 + |\chi_1|^2 = N.$$

The authors then impose (i) resonant rotating-wave terms, (ii) a slowly varying 2D Fourier envelope, (iii) dipole approximation $e^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}} \simeq 1$, (iv) identical collective coupling. Write

$$J_{mn} = \int \varphi_m^* \frac{\mathrm{i}\partial_3}{m} \varphi_n \mathrm{d}^3x, \quad g = \frac{eJ_{10}}{2}, \quad \mathcal{A} = \int_{\omega_\mathbf{k}=\Omega} \alpha_{3,\mathbf{k}}.$$

The resonant 2D shell is assigned the measure

$$\int_{\omega_{\mathbf{k}}=\Omega} \mathrm{d}\mu(\mathbf{k}) = \frac{\Omega^2}{4\pi}. \qquad \text{(A29; chosen shell convention)}$$

$$\dot{\chi}_0 = -\frac{\mathrm{i}e}{2} J_{01}\chi_1 \mathcal{A}^*, \qquad \text{(A30)}$$

$$\dot{\chi}_1 = -\frac{\mathrm{i}e}{2} J_{10}\boxed{\chi_1}\mathcal{A}, \qquad \text{(A31 as printed)}$$

$$\dot{\mathcal{A}} = -\frac{\mathrm{i}e\Omega}{4\pi L_3} J_{01}\chi_0^*\chi_1. \qquad \text{(A32)}$$

But (A17) contains χ_0 in the excited-state equation, and the claimed next equation (A34) also uses χ_0:

$$\dot{\chi}_0 = g\mathscr{A}\chi_1, \quad \boxed{\dot{\chi}_1 = -g\mathscr{A}\chi_0}, \quad \dot{\mathscr{A}} = \frac{g\Omega}{4\pi L_3}(2\chi_0^*\chi_1), \qquad \mathcal{A} \mapsto -\mathrm{i}\mathscr{A}. \qquad \text{(A33–A35)}$$

$\boxed{\text{(A31) cannot produce (A34) as printed; } \chi_1 \text{ must be } \chi_0.}$ [FAIL AS WRITTEN]

This may be a single-symbol typographical error; it must still be repaired rather than silently propagated.

4. The Bloch-sphere identity requires a phase condition

Source: A33–A40; p.17 Under the corrected (A34):

$$Z = |\chi_1|^2 - |\chi_0|^2, \qquad R = 2\chi_0^*\chi_1.$$

$$\dot{Z} = -2g\mathscr{A}R, \quad \dot{R} = 2g\mathscr{A}Z, \quad \dot{\mathscr{A}} = \frac{g\Omega}{4\pi L_3}R, \qquad \text{(A36–A38)}$$

provided R and $\mathscr{A}$ are real. The general identity is instead

$$\begin{aligned} N^2 &= (|\chi_0|^2 + |\chi_1|^2)^2 \\ &= \underbrace{(|\chi_1|^2 - |\chi_0|^2)^2}_{Z^2} + 4|\chi_0^*\chi_1|^2 \\ &= Z^2 + \underbrace{(2\Re\chi_0^*\chi_1)^2}_{R_\Re^2} + \underbrace{(2\Im\chi_0^*\chi_1)^2}_{R_\Im^2}. \end{aligned}$$

Hence $Z^2 + R_\Re^2 = N^2$ only when $\Im(\chi_0^*\chi_1) = 0$.

$$\chi_0 = \sqrt{N/2}, \quad \chi_1 = \mathrm{i}\sqrt{N/2} \Longrightarrow Z = 0, \quad R_\Re = 0, \quad N^2 > 0.$$

The paper's substitution $Z = N\cos\theta$, $R = N\sin\theta$ therefore selects the real-coherence sector. [CONDITIONAL]

Owed: State the permitted relative phase/initial data and preserve it under the selected real-field dynamics.

5. Pulse solution: reproduce it, then test its approximations

Source: A38–A47; main text (1)–(2) Paper first derives $\dot{\mathscr{A}} = \frac{g\Omega N}{4\pi L_3}\sin\theta$. It **then adds** the loss term:

$$\boxed{\dot{\mathscr{A}} + \frac{1}{L_3}\mathscr{A} = \frac{g\Omega N}{4\pi L_3}\sin\theta.} \qquad \text{(A42: introduced loss)}$$

Take $|L_3\dot{\mathscr{A}}| \ll |\mathscr{A}|$:

$$\mathscr{A} \simeq \frac{g\Omega N}{4\pi}\sin\theta, \qquad \dot{\theta} = 2g\mathscr{A} = \frac{g^2\Omega N}{2\pi}\sin\theta =: \frac{1}{\tau_R}\sin\theta.$$

$$\int \frac{\mathrm{d}\theta}{\sin\theta} = \int \frac{\mathrm{d}t}{\tau_R} \Longrightarrow \ln\tan\frac{\theta}{2} = \frac{t}{\tau_R} + \ln\tan\frac{\theta_0}{2},$$

$$\tau_R = \frac{2\pi}{g^2\Omega N}, \quad \tau_0 = -\tau_R \ln\tan\frac{\theta_0}{2}, \quad \theta = 2\arctan e^{(t-\tau_0)/\tau_R}.$$

$$\boxed{\mathscr{A}(t) = \frac{g\Omega N}{4\pi}\operatorname{sech}\frac{t-\tau_0}{\tau_R}, \quad |E_3| \simeq \Omega|\mathscr{A}| = \frac{g\Omega^2 N}{4\pi}\operatorname{sech}\frac{t-\tau_0}{\tau_R}.} \text{ [CHECK]} \qquad \text{(A45–A47)}$$

But the claimed "spontaneous" burst is not generated from an exactly unseeded classical initial state:

$$\theta_0 = 0, \quad \mathscr{A}(0) = 0 \Longrightarrow \dot{\theta}(0) = 0, \quad \boxed{\theta(t) \equiv 0, \quad \mathscr{A}(t) \equiv 0.}$$

$$\lim_{\theta_0 \to 0^+} \tau_0 = +\infty.$$

An initial coherence/seed or quantum-noise process is required. No stochastic vacuum source is propagated through (A33)–(A47). [NOT SHOWN]

Self-consistency of the adiabatic step:

$$\frac{|\dot{\mathscr{A}}|}{|\mathscr{A}|/L_3} = \frac{L_3}{\tau_R}\left|\tanh\frac{t-\tau_0}{\tau_R}\right| \quad \Longrightarrow \quad \boxed{\frac{g^2\Omega N L_3}{2\pi} \ll 1} \quad \text{away from the pulse center.}$$

The formal limit $N \to \infty$ at fixed g, Ω, L_3 violates the condition used to derive the N-dependent pulse. [CONDITIONAL]

Owed: A nonzero physical seed/noise source; values of g, Ω, N, L_3; an error estimate for the longitudinal-field and adiabatic truncations.

6. Holography: exact arithmetic and unstated spectral condition

Source: A48–A52; pp.18–19 Set $\mu^2(x) = e^2|\phi(x)|^2/m$; the paper **adds** M^2 and an external drive $\tilde{A}_i$:

$$\boxed{(\Box + \mu^2 + M^2)A_i = M^2\tilde{A}_i + O(\partial\phi, \partial\phi^*), \quad \tilde{A}_i = b\,e^{\mathrm{i}(\omega t - \mathbf{k}\cdot\mathbf{x})}.} \qquad \text{(A48)}$$

For spatially constant μ, try $A_i = Ce^{\mathrm{i}(\omega t - \mathbf{k}\cdot\mathbf{x})}$:

$$[\underbrace{-\omega^2 + |\mathbf{k}|^2}_{\Box \text{ contribution}} + \mu^2 + M^2]C = M^2 b,$$

$$\boxed{C = \frac{M^2 b}{M^2 + \mu^2 + |\mathbf{k}|^2 - \omega^2}.}$$

Paper (A49): $C = \frac{M^2 b}{M^2+\mu^2}$. Thus (A49) requires $\omega^2 = |\mathbf{k}|^2$ *and* negligible gradients of μ; its series also requires $|\mu^2| \ll M^2$. [CONDITIONAL]

$$\tilde{A}_i = r(x^1, x^2), \quad |\phi|^2 = \kappa |r + o|^2 \quad (\kappa > 0 \text{ supplied}). \tag{A50–A51}$$

$$|r + o|^2 = |r|^2 + |o|^2 + r^* o + r o^*,$$
$$r|r + o|^2 = r|r|^2 + r|o|^2 + \boxed{|r|^2 o} + \boxed{r^2 o^*}.$$

A desired object term $|r|^2 o$ exists, but so do the twin image $r^2 o^*$ and other orders; optical selection/readout is not specified. [CONDITIONAL]

Source-symbol check: (A50) defines $\tilde{A}_i = r$ as *input*; (A49) solves A_i as *response*. (A52) prints $\tilde{A}_i \propto r|r + o|^2$, which appears to exchange input and response. Algebra yields a term in A_i, not in a previously fixed $\tilde{A}_i$. [FAIL AS WRITTEN]

Owed: The origin of o; a physical write/read channel; reference beam; separation of image orders; corrected input/output symbol in (A52).

7. Capacity: no QED required for the quoted number

Source: main text (3)–(5), p.6

$$\frac{C}{S} = \frac{4 n_r d}{\lambda^3}, \quad n_r = 1.3, \quad d = 2\,\text{mm}, \quad \lambda = 500\,\text{nm} = 5 \times 10^{-4}\,\text{mm}.$$

$$\frac{C}{S} = \frac{4(1.3)(2)}{(5 \times 10^{-4})^3} = 8.32 \times 10^{10}\ \text{bit/mm}^2.$$

$$S = 2500\,\text{cm}^2 = 2.5 \times 10^5\,\text{mm}^2, \quad C = 2.08 \times 10^{16}\ \text{bit} = \boxed{2.60\,\text{PB}}.$$

$$C_{\text{syn}} = \frac{(250 \times 10^{12}\ \text{synapses})(1\ \text{bit/synapse})}{8} = \boxed{31.25\,\text{TB}}.$$

The paper rounds to 2.5 PB and 30 TB. The arithmetic is consistent at this precision. [CHECK]

Source-to-capacity gap:

$\boxed{\lambda_{\text{emit}} = \dfrac{2\pi}{\Omega} \quad (c = \hbar = 1)}$, Ω, g, N never numerically determined for the proposed material.

The capacity calculation chooses $\lambda = 500$ nm; it does not compute that wavelength from the superradiance model. For scale *only*, $E_\gamma = hc/(500\,\text{nm}) = 2.48$ eV. If one *separately assumed* a 4 meV mode (a scale mentioned in the introduction), the wavelength would be $\approx 310\,\mu$m and the same formula would instead yield ≈ 11 MB. This is a sensitivity illustration, *not* a claim that the paper identified Ω with 4 meV.

8. Computer-generated hologram equations

Source: main text (6)–(9), p.6

$$|\nu_1 - (n_1 + P_{n_1 n_2})\delta\nu| \le \frac{C}{2}\delta\nu, \qquad |\nu_2 - n_2\delta\nu| \le \frac{W_{n_1 n_2}}{2}\delta\nu,$$

$$\widetilde{\mathcal{U}}(n_1/X^1, n_2/X^2) \propto \mathcal{B}_{n_1 n_2} e^{\mathrm{i}\Phi_{n_1 n_2}}, \qquad W_{n_1 n_2} = \mathcal{B}_{n_1 n_2}, \qquad 2\pi\mathcal{N} P_{n_1 n_2} = \Phi_{n_1 n_2}.$$

A prescribed Fourier amplitude/phase is encoded into geometric width/offset. This is an algorithm, not a derivation of the source image $\mathcal{U}$ from neural dynamics. [CHECK] (conditional construction).

9. 2PI branch: retain quantum terms until the truncation

Source: A53–A75, pp.19–21

$$\mathcal{L}_C = -\tfrac{1}{4}F[A+a]^2 - \frac{(\partial_\mu a^\mu)^2}{2\xi} + \phi^*\left(\mathrm{i}\partial_t + e(A^0 + a^0) + \frac{[\nabla - \mathrm{i}e(\mathbf{A}+\mathbf{a})]^2}{2m}\right)\phi. \tag{A53}$$

$$\phi \mapsto e^{\mathrm{i}\alpha}\phi, \quad A_\mu \mapsto A_\mu + e^{-1}\partial_\mu\alpha, \quad a_\mu \mapsto a_\mu; \qquad a^0 = 0, \quad \xi = 1. \tag{A54–A55}$$

Using $\bar{\phi} = \langle\phi\rangle$, $\Delta(x,y) = \langle T_C \delta\phi(x)\delta\phi^*(y)\rangle$, and $D_{ij}(x,y) = \langle T_C a_i(x) a_j(y)\rangle$, (A56) is a 2PI functional of the form

$$\Gamma_{\mathrm{2PI}} = S[\bar{\phi}, A] + \frac{\mathrm{i}}{2}\mathrm{Tr}\ln D^{-1} + \frac{1}{2}\mathrm{Tr}(\mathrm{i}D_0^{-1}D) + \mathrm{i}\mathrm{Tr}\ln\Delta^{-1} + \mathrm{Tr}(\mathrm{i}\Delta_0^{-1}\Delta) + \frac{1}{2}\Gamma_2[A, \bar{\phi}, \Delta, D].$$

Its stationarity equations (A62)–(A64) contain

$$\frac{\delta\Gamma_2}{\delta A_i}, \quad \frac{\delta\Gamma_2}{\delta\bar{\phi}^{(*)}}, \quad \Delta^{11}(x,x), \quad D_{ii}^{11}(x,x).$$

The concrete Γ_2 functional / loop truncation and propagators are not evaluated numerically in the Section 4 simulation. [NOT SHOWN]

Explicit phenomenological replacement in (A75):

$$-\frac{1}{2}\frac{\delta\Gamma_2}{\delta A_i} \simeq -M^2\left(A_i - \frac{1}{e}\partial_i\beta\right) - \gamma\,\partial_t\left(A_i - \frac{1}{e}\partial_i\beta\right) + O(\partial^2 A).$$

M^2 and γ are not computed from an evaluated Γ_2 for the proposed tissue; they are set as parameters. The output equations are therefore effective mean-field equations with imposed local mass/damping, not a numerical solution of the displayed full 2PI equations. [CONDITIONAL]

10. A direct sign counterexample in the phase equation

Source: A63, A69–A70, A81; pp.20–22 Let $\bar{\phi} = Re^{\mathrm{i}\beta}$, $R > 0$; omit the explicitly neglected quantum terms as the paper does. Start from

$$\left[\mathrm{i}\partial_t + eA_0 + \frac{(\nabla - \mathrm{i}e\mathbf{A})^2}{2m}\right]\bar{\phi} = 0.$$

Real part divided by R:

$$-\partial_t\beta + eA_0 + \frac{\nabla^2 R}{2mR} - \frac{|\nabla\beta - e\mathbf{A}|^2}{2m} = 0.$$

Thus

$$\boxed{A_0 - \frac{1}{e}\partial_t\beta = -\frac{\nabla^2 R}{2meR} + \frac{e}{2m}\left|\mathbf{A} - \frac{1}{e}\nabla\beta\right|^2.} \qquad \text{(computed)}$$

Paper (A70) prints a **minus** on the final squared term. [FAIL AS WRITTEN]

An independent substitution detects the error:

$$R = R_0 > 0, \quad \mathbf{A} = 0, \quad A_0 = 0, \quad \beta = \mathbf{k}\cdot\mathbf{x} - \frac{|\mathbf{k}|^2}{2m}t$$

solves the free Schrödinger equation. But the printed (A70) gives

$$\underbrace{A_0 - e^{-1}\partial_t\beta}_{+|\mathbf{k}|^2/(2me)} \overset{\text{(A70)}}{=} \underbrace{-\frac{e}{2m}|0 - e^{-1}\mathbf{k}|^2}_{-|\mathbf{k}|^2/(2me)}$$

for nonzero $\mathbf{k}$: contradiction. The paper's later (A81) has the *plus* sign, matching the independent derivation when $\beta = 0$. [FAIL AS WRITTEN]

Owed: Correct (A70), then explicitly recheck each subsequent substitution and gauge-invariant identity.

11. Two-dimensional driven-layer PDE: extract the actual simulation

Source: A76–A91; Section 4, pp.7–9, 22–23 Take the paper's $A_3 = 0$, $\partial_3 = 0$ and gauge $\beta = 0$; set $\rho = |\bar{\phi}|^2$. Equations (A76)–(A81) are

$$\begin{aligned}
\dot{E}_1 &= \partial_2^2 A_1 - \partial_1\partial_2 A_2 - \left(\frac{e^2\rho}{m} + M^2\right)A_1 - \gamma(E_1 + \partial_1 A_0) + vu_1,\\
\dot{A}_1 &= E_1 + \partial_1 A_0,\\
\dot{E}_2 &= \partial_1^2 A_2 - \partial_1\partial_2 A_1 - \left(\frac{e^2\rho}{m} + M^2\right)A_2 - \gamma(E_2 + \partial_2 A_0) + vu_2,\\
\dot{A}_2 &= E_2 + \partial_2 A_0,\\
\dot{\rho} &= \frac{e}{m}\left[\partial_1(\rho A_1) + \partial_2(\rho A_2)\right],\\
A_0 &= -\frac{(\partial_1^2 + \partial_2^2)\sqrt{\rho}}{2me\sqrt{\rho}} + \frac{e}{2m}(A_1^2 + A_2^2).
\end{aligned}$$

The nth layer (A82)–(A87) takes

$$\boxed{u_i^{(n)} = A_i^{(n-1)} + A_i^{(n+1)}, \qquad n = 1, \ldots, N,}$$

with the same local PDE repeated in each layer. No z-dependent Maxwell boundary-value problem is solved between layers.

Static inverse input (A88)–(A89):

$$\boxed{u_1^{(n-1)} = \frac{-\partial_2^2 u_1^{(n)} + \partial_1\partial_2 u_2^{(n)} + M^2 u_1^{(n)}}{v} - u_1^{(n+1)},}$$

$$\boxed{u_2^{(n-1)} = \frac{-\partial_1^2 u_2^{(n)} + \partial_1\partial_2 u_1^{(n)} + M^2 u_2^{(n)}}{v} - u_2^{(n+1)}.}$$

In deriving these, the paper explicitly neglects $\dot{E}_i, \gamma(E_i + \partial_i A_0)$ and *also* $e^2\rho A_i/m$. At its own input values $e = 0.3, m = 1, \rho(0) = 1, M^2 = 2$,

$$\frac{e^2\rho}{m} = 0.09, \qquad \frac{e^2\rho/m}{M^2} = \boxed{4.5\%}.$$

An omission is a quantitative approximation; an error bound for the *final driven output* is not supplied. [CONDITIONAL]

The target is piecewise linear, not globally twice differentiable. For (10), with $N_s = 64$, $N_1 = 30$, $f(n) = u_1^{(N)}(n)$,

$$f'(n) = \begin{cases} 0.1/(N_s - N_1), & n < -N_1, \\ -0.1/N_1, & -N_1 < n < N_1, \\ 0.1/(N_s - N_1), & n > N_1. \end{cases}$$

Hence in the distributional sense

$$f''(n) = \delta_{n=N_1}\Delta s - \delta_{n=-N_1}\Delta s, \quad \Delta s = 0.1\left(\frac{1}{30} + \frac{1}{34}\right) = 0.0062745\ldots$$

where $\delta_{n=a}$ denotes a Dirac delta at the corner (the discrete second difference is likewise nonzero at grid junctions). Similarly for (11) with $N_2 = 20$,

$$\Delta s_2 = 0.1\left(\frac{1}{20} + \frac{1}{44}\right) = 0.0072727\ldots$$

The paper's assertion that second derivatives vanish for the polygonal targets holds only *between* corners, not on the full grid. [FAIL AS WRITTEN] (if treated as a global identity).

The chosen input, not a spontaneous memory write:

$$A_i^{(0)}(t, \mathbf{x}) = u_i^{(0)}(\mathbf{x})e^{-\gamma_2 t}, \quad \frac{\gamma_2}{m} = 0.01, \quad N = 4, \quad \frac{v}{m}^2 = 0.8, \quad \frac{\gamma}{m} = 0.2, \quad \frac{M^2}{m^2} = 2.$$

$$\text{Input amplification scale } \sim 0.1\left(\frac{M^2}{v}\right)^4 = 0.1\left(\frac{2}{0.8}\right)^4 = 3.90625.$$

The report's $\sim 21\%$ density reduction at $mt = 200$ is a *numerical output* for those inputs and boundary conditions, not an observed biological memory. [CHECK] (report of simulation only).

Owed: Source code or numerical tables, derivative stencil, convergence and stability checks, and any physical calibration $m, \sigma, \varepsilon, \mu$.

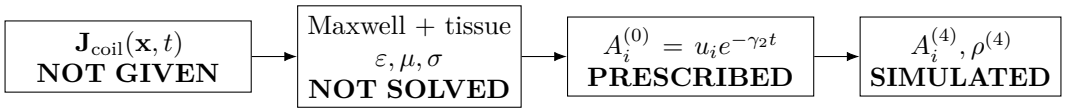
12. Field generation: coils, currents, Gauss, boundaries

Source: Fig. 1; Fig. 3; A76–A91 The paper supplies $A_i^{(0)}$ directly. It does **not** give a physical coil, a current source or a circuit equation. For a real electromagnetic driver, in SI,

$$\nabla \cdot \mathbf{D} = \rho_f, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{H} = \mathbf{J}_f + \partial_t \mathbf{D}, \quad \nabla \times \mathbf{E} = -\partial_t \mathbf{B},$$

$$\mathbf{D} = \varepsilon(\mathbf{x}, \omega)\mathbf{E}, \quad \mathbf{B} = \mu(\mathbf{x}, \omega)\mathbf{H}, \quad \mathbf{J}_{\text{conduction}} = \sigma(\mathbf{x}, \omega)\mathbf{E},$$

plus interface conditions for $\mathbf{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1)$, $\mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1)$, $\mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1)$, $\mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1)$.



For even an ideal circular coil, $B_z(0) = \mu_0 N_c I/(2R_c)$ requires turns N_c, current I, radius R_c; none is specified. The link $\mathbf{J}_{\text{coil}} \mapsto A_i^{(0)}$ is therefore NOT CALCULATED.

Closed vs open energy bookkeeping. For $\ddot{A} - \nabla^2 A + M_{\text{eff}}^2 A + \gamma \dot{A} = f$,

$$H = \int \left(\frac{1}{2}\dot{A}^2 + \frac{1}{2}|\nabla A|^2 + \frac{1}{2} M_{\text{eff}}^2 A^2 \right) \mathrm{d}^d x, \qquad \dot{H} = -\gamma \int \dot{A}^2 \mathrm{d}^d x + \int f \dot{A} \mathrm{d}^d x$$

for constant M_{eff}. The paper's earlier conserved-energy identity (A73)–(A74) does not apply unchanged to its reduced damped, driven sector without accounting for reservoir/source energy. [CONDITIONAL]

13. Independent algebraic verdict, by equation range

Source	Result	What must be shown or corrected
(A1)–(A7)	[FAIL AS WRITTEN]	$A^0 = 0$ and $\nabla \cdot \mathbf{A} = 0$ contradict $J_0 = -e\|\phi\|^2$, $N > 0$.
(A8)–(A11)	[CONDITIONAL]	Fixed eigenbasis despite time-dependent $\mathbf{A}$; longitudinal averaging and current geometry.
(A12)–(A29)	[CONDITIONAL]	RWA, dipole, mode-density and envelope approximations admitted; no material parameter extraction.
(A30)–(A35)	[FAIL AS WRITTEN]	(A31) uses χ_1 as printed; (A34) requires χ_0.
(A36)–(A40)	[CONDITIONAL]	$Z^2 + R^2 = N^2$ requires real relative coherence.
(A42)–(A47)	[CHECK] conditional	sech derived from seeded adiabatic Bloch dynamics; no spontaneous seed supplied.
(A48)–(A49)	[CONDITIONAL]	Added M^2 drive; omitted $-\omega^2 + \|\mathbf{k}\|^2$ requires on-shell input.
(A50)–(A52)	[FAIL AS WRITTEN]	Hologram density assumed; input $\tilde{A}_i$ vs response A_i confused in (A52).
(3)–(5)	[CHECK] arithmetic	2.60 PB vs 31.25 TB with chosen λ, n, d, S; optical write/read not established.
(6)–(9)	[CHECK] conditional	A standard supplied binary encoding algorithm; no biological encoding derivation.
(A53)–(A68)	[CONDITIONAL]	2PI notation genuine; Γ_2 and Green functions not evaluated for Section 4 output.
(A69)–(A81)	[FAIL AS WRITTEN]	(A70) has wrong sign in squared phase-gradient; (A81) has opposite sign.
(A75)–(A91)	[CONDITIONAL]	Quantum terms modeled/neglected, effective mass/damping, prescribed forcing, unbounded omitted terms.
(10)–(11), (A88)–(A89)	[CONDITIONAL]	Inverse-designed piecewise linear target; second differences nonzero at corners; finite coupling discarded.
Figs.5–6	[NOT SHOWN] (biological)	Numerical output is not an experiment in a brain; no unique quantum-only effect calculated.

14. Residual calculation requests (no inferred biology)

$\mathcal{R}_1$: consistent gauge + nonzero current J^μ,
$\mathcal{R}_2$: material-specific $(m, e, U, \Omega, J_{01}, N, L_3)$,
$\mathcal{R}_3$: RWA/dipole/adiabatic error bounds, physical source seed,
$\mathcal{R}_4$: derive λ from Ω rather than choosing it,
$\mathcal{R}_5$: derive M^2, γ from specified Γ_2,
$\mathcal{R}_6$: correct (A31), (A70), (A52), and retest dependents,
$\mathcal{R}_7$: source current + tissue Maxwell boundary solution,
$\mathcal{R}_8$: compare with a matched classical driven-field model,
$\mathcal{R}_9$: one measured write/store/read/retention result in tissue.

Supplied model + chosen approximations
$\Rightarrow$ conditional pulse, optical algebra, capacity, simulated pattern;
the equations do not establish biological memory,
a uniquely quantum effect, or consciousness.

Source inspected: Kobe University repository version-of-record PDF, journal pp.5–23 (equations). Each counterexample and inference tagged "computed" is independent analysis; authors' displayed equations are identified by their original labels.